

City Flooding (manhole)

William is playing a city simulation game: the main objective is to save the mayor's life in the event of natural disasters. . . and the weather forecast just warned William about an imminent flooding!

The city can be seen as a straight line: there are N positions numbered from 0 to $N - 1$. The mayor's house is in position K . During the flooding, L liters of water will flow from the left corner to position 0 first, then to position 1 and so on.

In order to stop the water from flooding the mayor's home, you need to place $N - 1$ manholes in every position except the one in which the house of the mayor is located. Luckily, you have exactly $N - 1$ manholes left in the game inventory, however, they are all different. Some are able to "subtract" more water than others: **no two manholes** will subtract **the same** amount!



For example, let's say that $L = 100$ liters of water will rain down during the flooding. Moreover, let's assume that the city has $N = 5$ total positions (numbered from 0 to 4), and that $K = 3$ is the position of the mayor's house. In this case, we would have $N - 1 = 4$ manholes in our inventory. Let's say that the 4 manholes are able to remove, respectively: 15, 80, 20, and 50 liters of water.

In this case, we can see that there are only 6 ways to arrange the manholes so that some water *will* hit the mayor's home (indicated with a ★). To be precise, $100 - (15 + 20 + 50) = 15$ liters of water will reach his home.

0	1	2	3	4
15	20	50	★	80
20	15	50	★	80
20	50	15	★	80
15	50	20	★	80
50	15	20	★	80
50	20	15	★	80

So, the number of *bad* manhole arrangements is 6. Since the total number of arrangements is $4! = 24$, we can deduce that the answer for this case is $24 - 6 = 18$.

Compute how many ways William can put the $N - 1$ manholes in such a way that will keep the mayor's house dry during the flooding.

📎 Among the attachments of this task you may find a template file `manhole.*` with a sample incomplete implementation.

Input

The first line contains a single integer: L . The second line contains two integers: N and K . The third line contains $N - 1$ integers $V_0, V_1, \dots, V_{N-1}$: the "capacity" of the manholes.

Output

You need to write a single line with an integer: the number of good manhole arrangements. Since this number can be huge, just print the remainder when dividing it by 10 007.

🔗 To execute the remainder operation di modulo you can use C/C++ operator `%`. In Pascal you can use operator `mod`.

For example, the remainder of the division of 5 by 3 can be computed as `5 % 3` in C/C++ and as `5 mod 3` in Pascal. Both will return 2.

Moreover, the remainder operation has a number of properties which can be very useful (e.g. to avoid *integer overflow* when working with very large numbers):

- $(A + B) \bmod M = (A \bmod M + B \bmod M) \bmod M$
- $(A \cdot B) \bmod M = (A \bmod M \cdot B \bmod M) \bmod M$

Constraints

- $1 \leq L \leq 1000$.
- $2 \leq N \leq 100$.
- $0 \leq K \leq N - 1$.
- $0 \leq V_i \leq 1000$ for each $i = 0 \dots N - 2$.
- All V_i values are different.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [15 points]: $K = N - 1$, that is, the mayor lives in the last position.
- **Subtask 3** [20 points]: $N \leq 8$.
- **Subtask 4** [30 points]: $N \leq 16$.
- **Subtask 5** [30 points]: No additional limitations.

Examples

input.txt	output.txt
100 5 3 15 80 20 50	18
100 10 5 34 30 16 9 18 14 27 10 15	5415

Explanation

The **first sample case** is the one described in the task statement.

In the **second sample case** there are 155 520 valid combinations. Since we are requested to print only the remainder of the division by 10 007, the answer is $155\,520 \% 10\,007 = 5415$.